2. Match the following:

| Group A<br>Exponential form | | Group B<br>Unit digit of the number | |
|---|---|---|---|
| (i) | $20^{10}$ | (a) | 6 |
| (ii) | $121^{11}$ | (b) | 4 |
| (iii) | $444^{41}$ | (c) | 0 |
| (iv) | $25^{100}$ | (d) | 1 |
| (v) | $716^{83}$ | (e) | 9 |
| (vi) | $729^{725}$ | (f) | 5 |

3. Find the unit digit of expanded form.

(i) $25^{23}$ (ii) $11^{10}$ (iii) $46^{15}$ (iv) $100^{12}$

(v) $29^{21}$ (vi) $19^{12}$ (vii) $24^{25}$ (viii) $34^{16}$

4. Find the unit digit of the following numeric expressions.

(i) $114^{20}+115^{21}+116^{22}$ (ii) $10000^{10000}+11111^{11111}$

### Objective type questions

5. Observe the equation $(10+y)^4 = 50625$ and find the value of $y$.

(i) 1 (ii) 5 (iii) 4 (iv) 0

6. The unit digit of $(32\times 65)^0$ is

(i) 2 (ii) 5 (iii) 0 (iv) 1

7. The unit digit of the numeric expression $10^{71}+10^{72}+10^{73}$ is

(i) 0 (ii) 3 (iii) 1 (iv) 2

## 3.5 Degree of Expression

Let us recall about algebraic expression which we have studied earlier.

### 3.5.1 Recap of Algebraic expression

We have learnt that while constructing an algebraic expression, we use mathematical operators like addition, subtraction, multiplication and division to combine variables and constants.

Now, we have learnt about exponents. Note that exponential notations also can be used in the construction of Algebraic expressions.

Let us recall some basic concepts about expressions.



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